AENL226 Power Electronics and power systems (Aug-Dec 2026)

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Lecture content

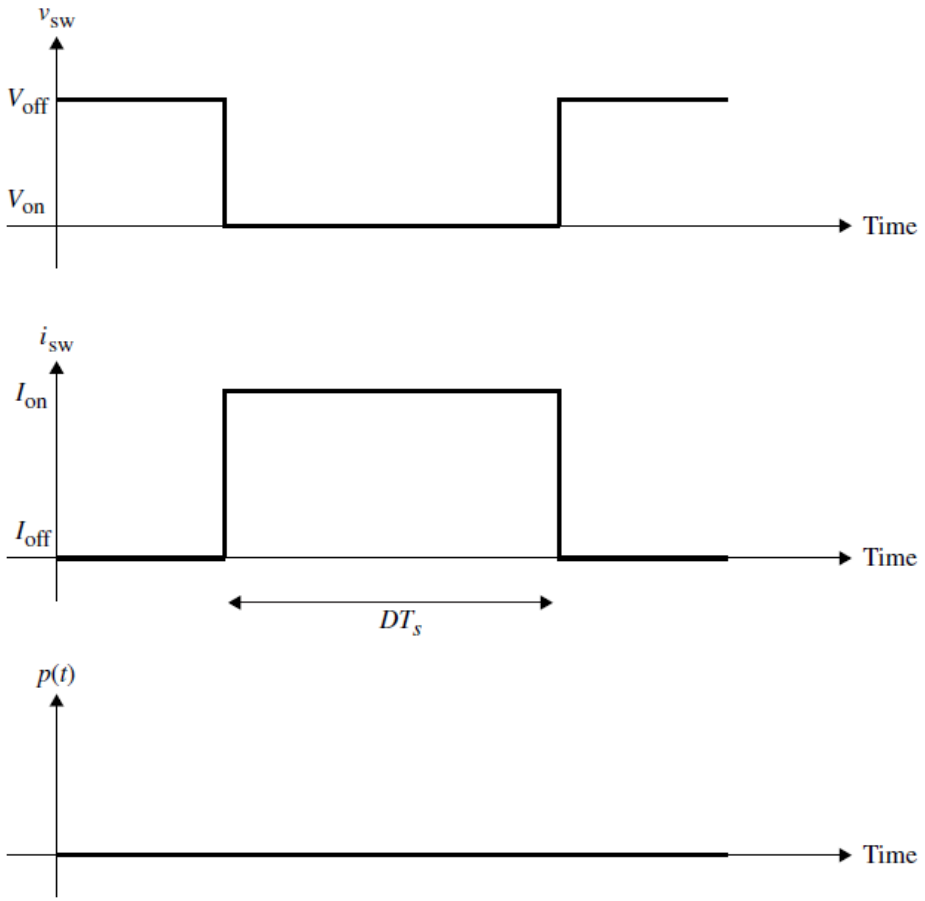
- Losses in power switches
- Heat sink and thermal design
- See also: Issa Batarseh, chapter 2
- NPTEL lectures

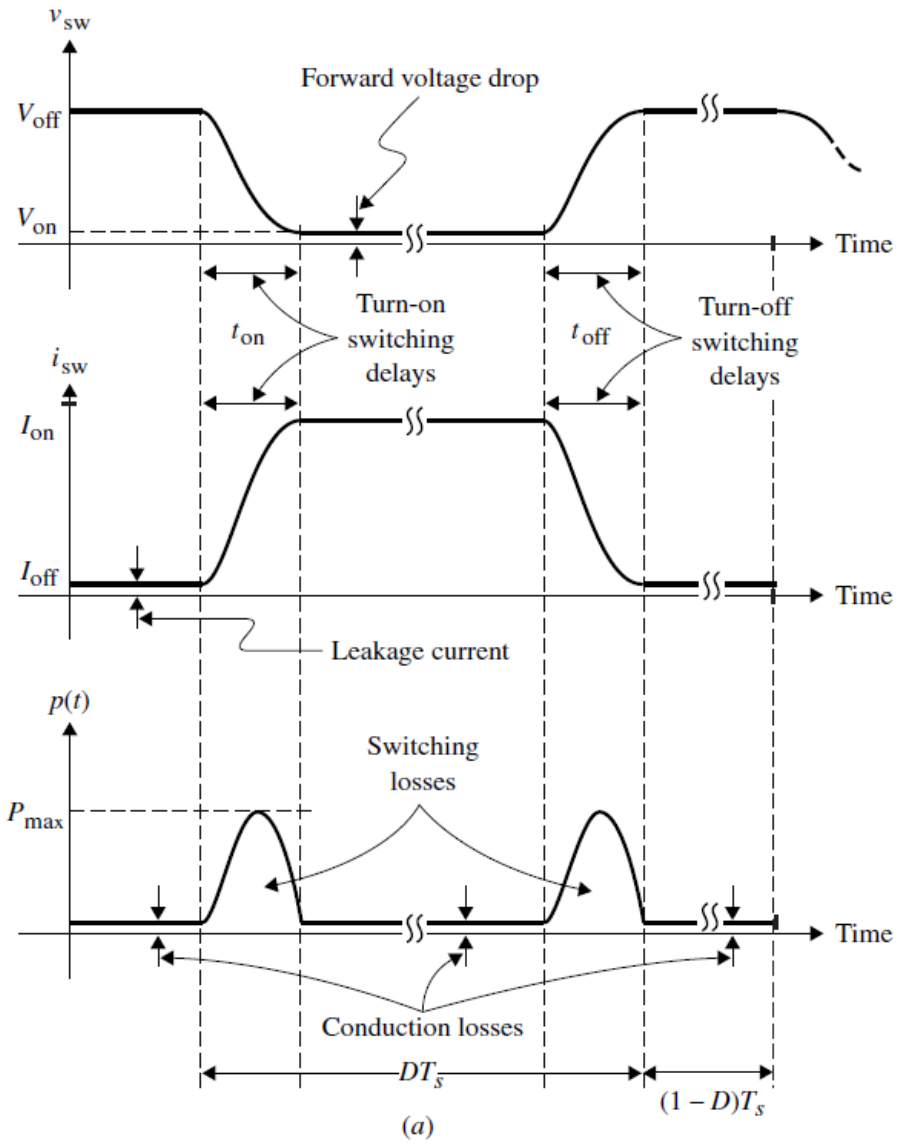
Losses in a practical switch

- There are mainly two types of losses in a practical switch- switching loss and conduction loss.
- Switching losses take place during turn on and turn off process of the device.
- Conduction loss takes place when the device is fully under conduction.

Practical switch timing diagram

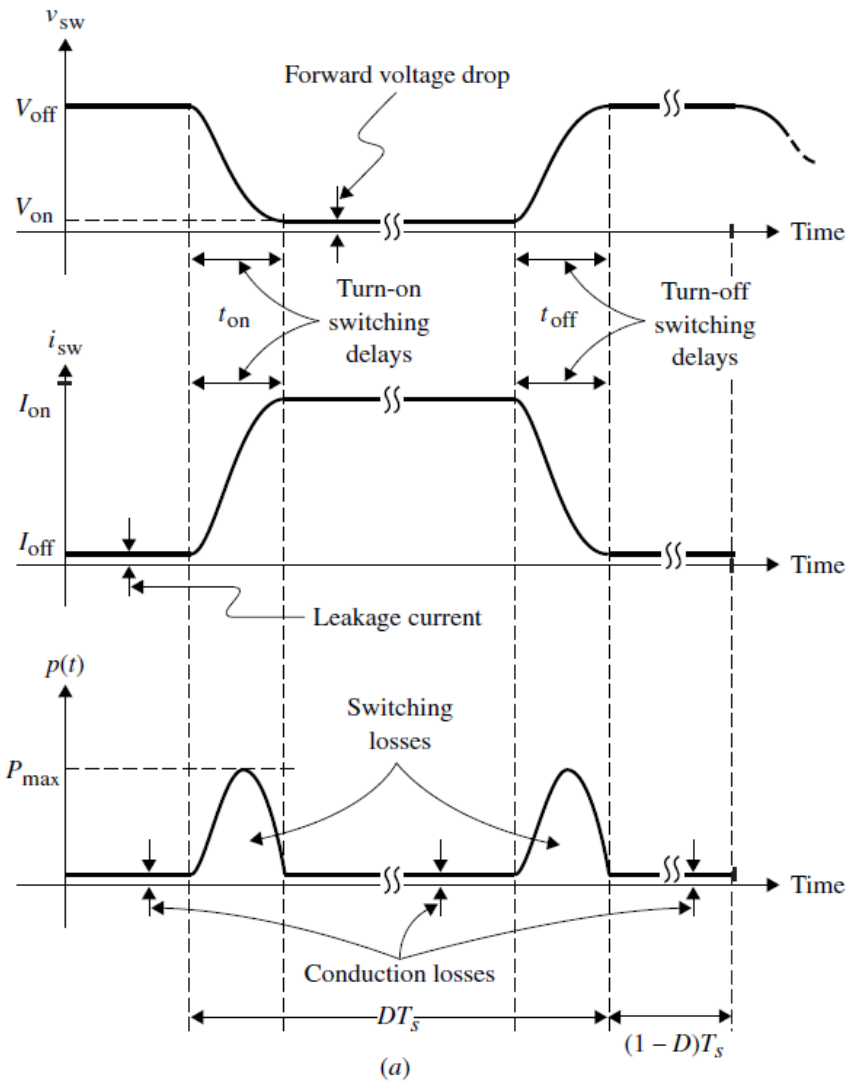
Ideal and practical waveforms





(a)

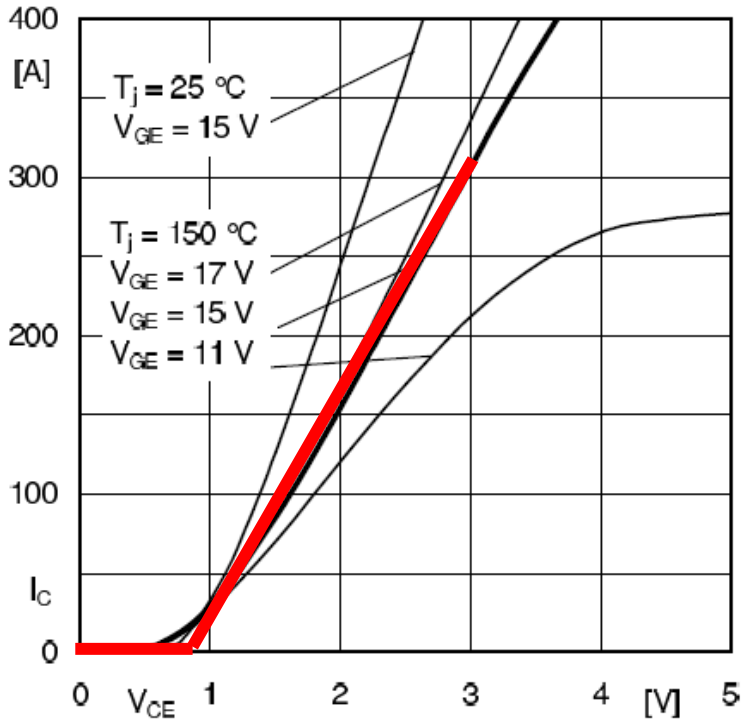
Practical switch timing diagram



(a)

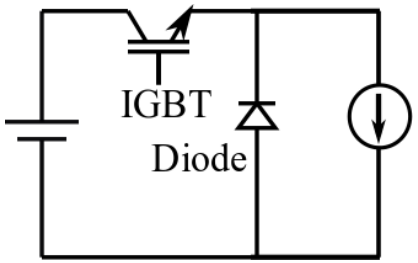
- The device has on state drop.
- The device takes finite turn-on and turn off times.

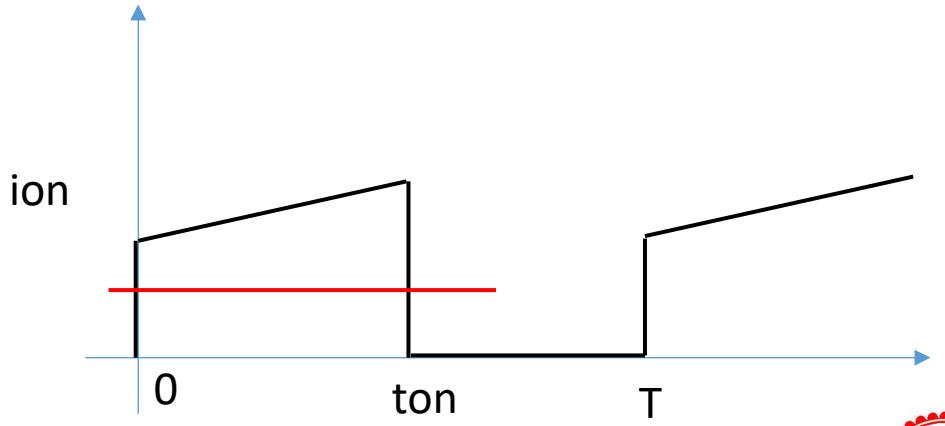
Conduction loss



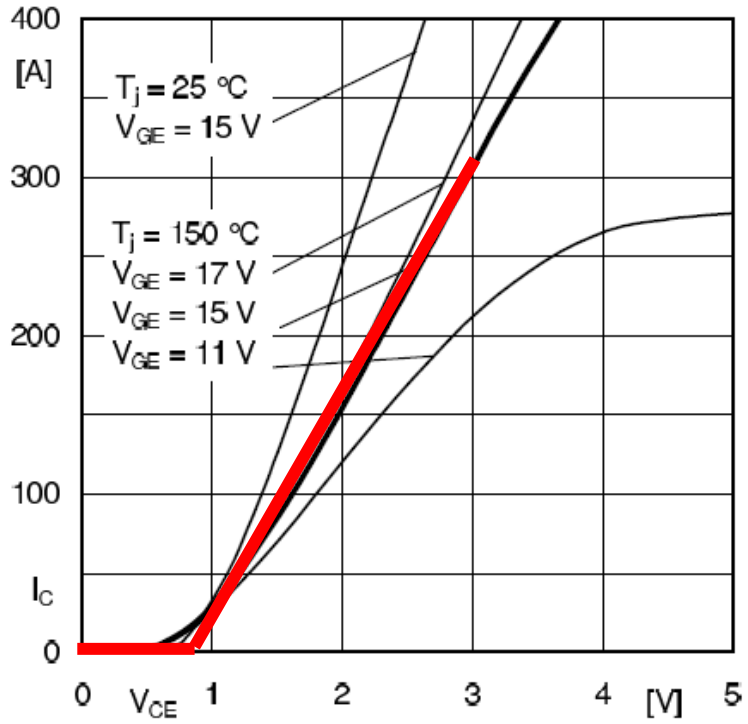
From SKM200GB12V datasheet

- Conduction loss in IGBT: $\frac{1}{\mathrm{T}}\int_0^{t_{on}} v_{ce}(t) i_{on}(t)\, dt$
- Suppose we use the IGBT in a buck converter and neglect the switch current ripple.
- Then, $p_{cond} = V_{ce0} i_{on} + R_{ce} i_{on}^2$
- $P_{cond} = \frac{1}{\mathrm{T}}\int_0^{t_{on}} p_{cond}\, dt = V_{ce0} I_{on_av} + R_{ce} I_{on_rms}^2$





Conduction loss



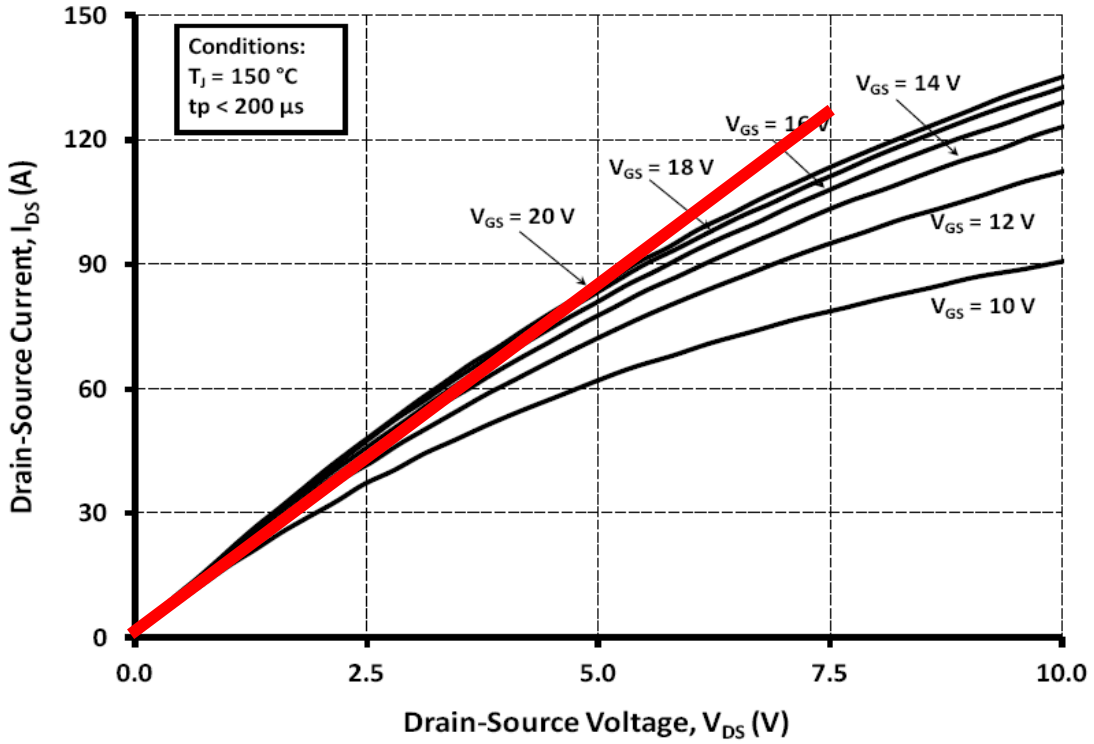
From SKM200GB12V datasheet

- From the figure, $R_{ce} = \frac{2V}{300A} = 6.66 \times 10^{-3}\Omega$
- For this IGBT if i_{on}= 150A and D=0.5, then (assuming negligible ripple)

$$P_{cond} = 0.9 \times 75 + 6.66 \times 10^{-3} \times 0.5 \times 150^2$$
$$= 67.5 + 75 = 142.5\ \text{W}$$

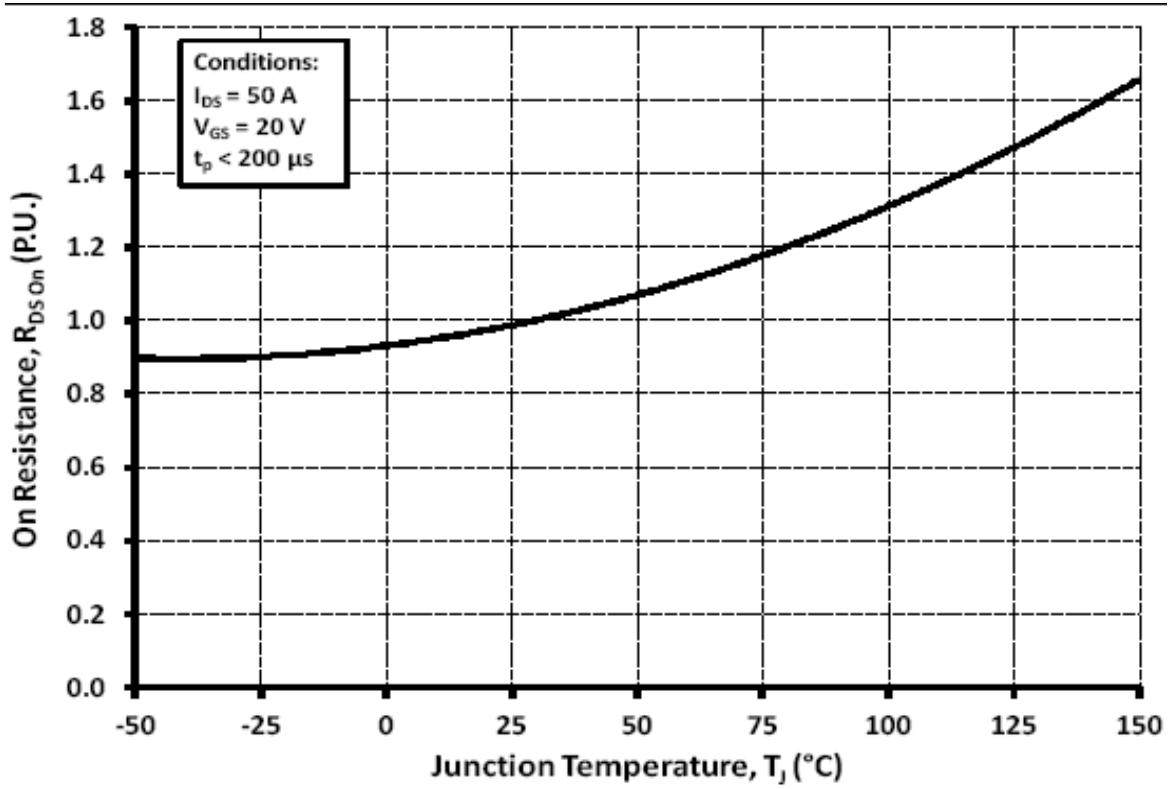
Conduction loss

- Conduction loss in MOSFET $= r_{dson} i_{on}^2$
- From MOSFET C2M002512 curve we get, r_{ds}= 45 mohm



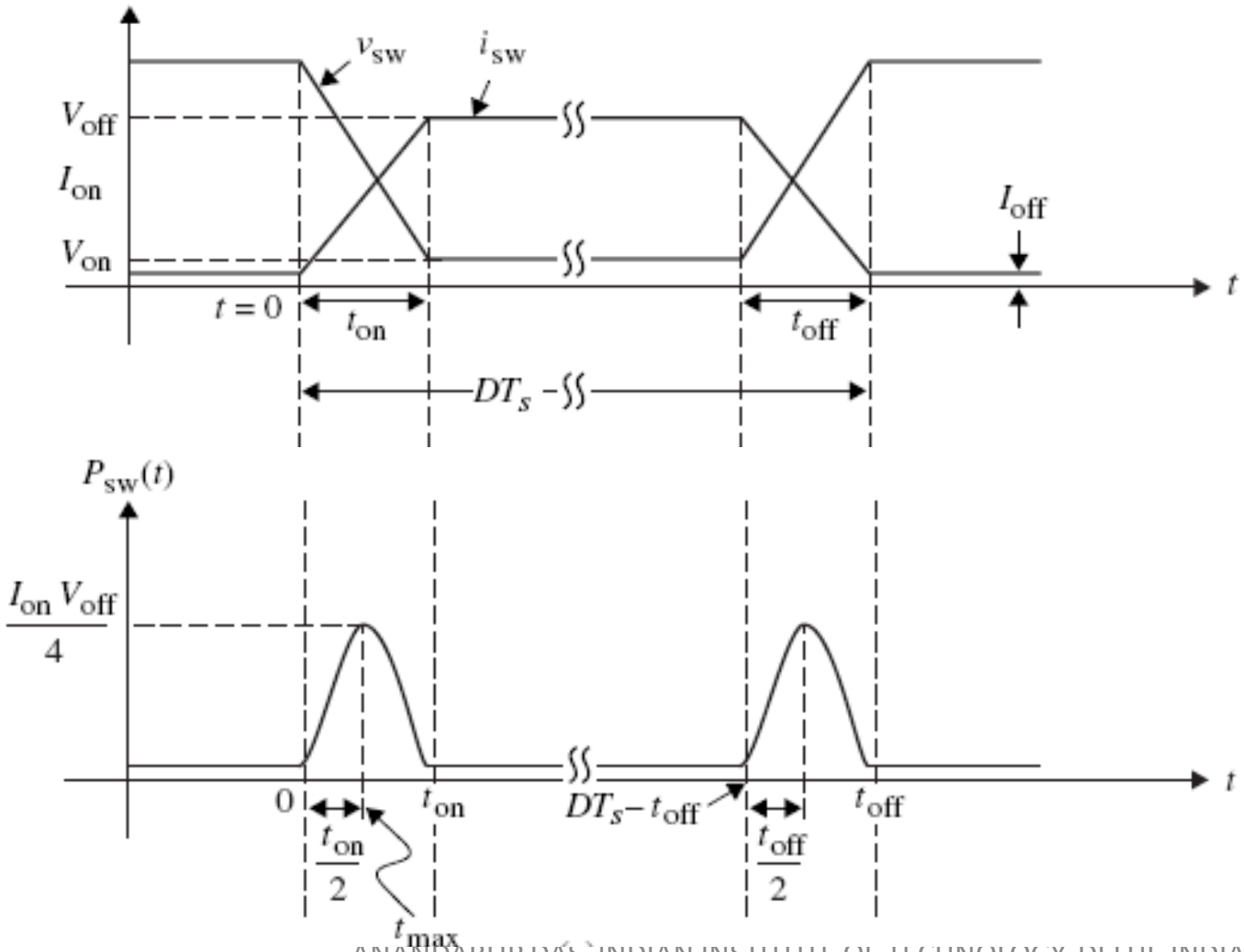
From C2M0025120D datasheet

Rds,on for MOSFET

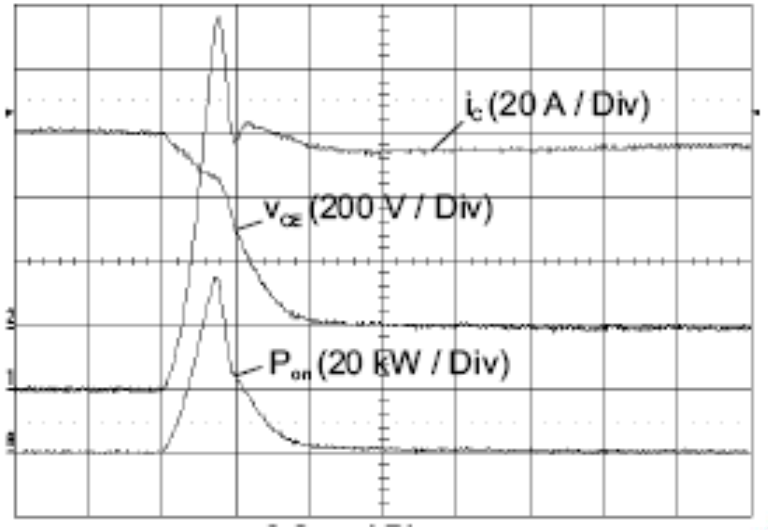


- Note that the on-state resistance of MOSFET changes with temperature.
- By observing the datasheet, many such information can be obtained.
- See the datasheet of IRF520NPbF, C2M0025120D and SKM200GB12V.

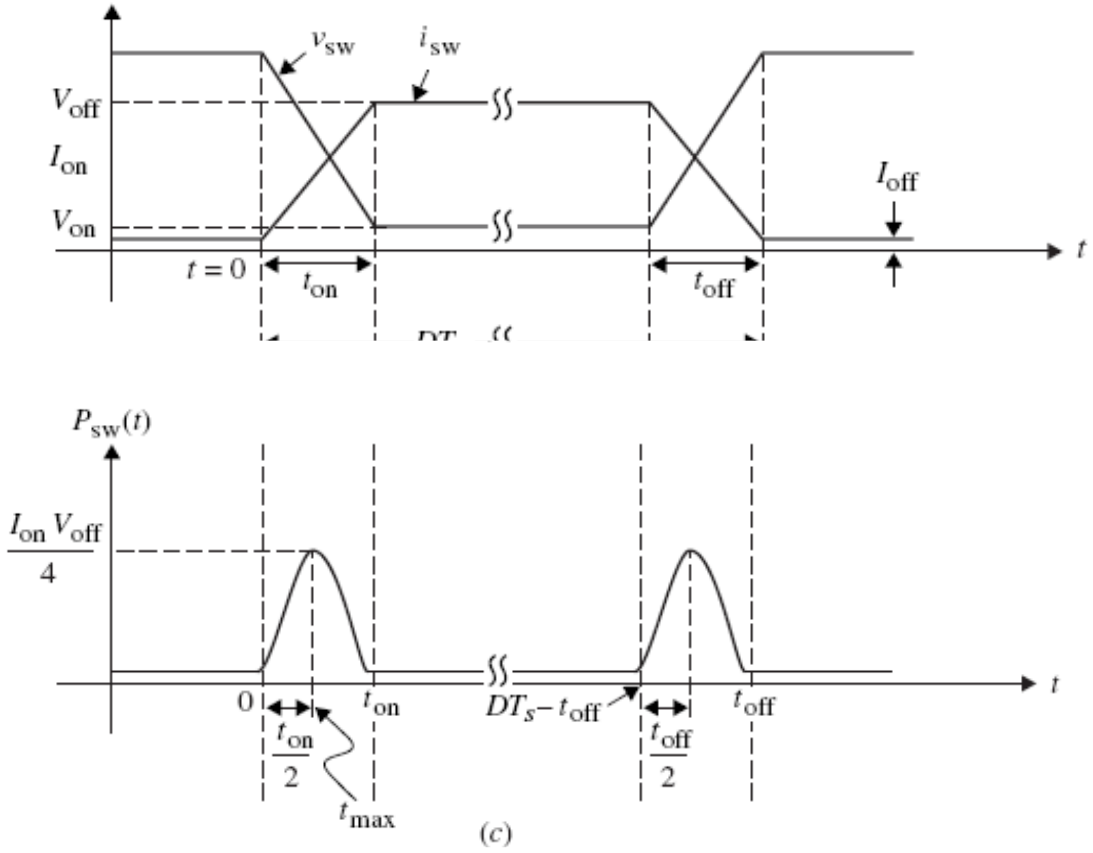
Switching loss



- The actual switching curve looks like below.
- It can be approximated by the left side curve.



Switching loss



- $i(t) = \frac{I_{on}}{t_{on}} t$ during turn on
- $i(t) = I_{on} - \frac{I_{on}}{t_{off}} t$ during turn off
- $v(t) = V_{off} - \frac{V_{off}}{t_{on}} t$ during turn on
- $v(t) = \frac{V_{off}}{t_{off}} t$ during turn off

- $E_{on} = \int_0^{t_{on}} i(t)v(t)dt = \frac{1}{6} V_{off} I_{on} t_{on}$
- $E_{off} = \int_0^{t_{off}} i(t)v(t)dt = \frac{1}{6} V_{off} I_{on} t_{off}$
- $P_{av} = \frac{1}{6T_S} V_{off} I_{on} (t_{on} + t_{off})$

$$= \frac{f_S}{6} V_{off} I_{on} (t_{on} + t_{off})$$

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Total Loss summary

- Every time the switch turns on or off, there is a pulse of energy dissipated.
- While the switch is fully turned on, there is a steady dissipation of energy.

- The total effect of these phenomena causes the temperature to rise inside the device.

Basic guideline

for hard switching:	MOSFET modules	low-voltage	up to 250 kHz
		high-voltage	up to 100 kHz
	IGBT modules	600 V	up to 30 kHz
		1200 V	up to 20 kHz
		1700 V	up to 10 kHz
		3300 V	up to 3 kHz
for soft switching:	MOSFET modules	low-voltage	up to 500 kHz
		high-voltage	up to 250 kHz
	IGBT modules		up to 150 kHz

- Taken from Semikron application note, see https://www.semikron.com/dl/service-support/downloads/download/semikron-application-manual-power-semiconductors-english-en-2015.pdf

Example

- A power MOSFET is used as a switch in a 48 V DC–DC converter operating at a switching frequency of 100 kHz. The converter supplies a resistive load drawing an average current of 10 A.
- The MOSFET has the following parameters:
  - Drain-source voltage when OFF: $V_{DS} = 48$ V
  - On-state resistance: $R_{DS(on)} = 40\ m\Omega$
  - Switching frequency: $f_s = 100$ kHz
  - Turn-on time: $t_{on} = 50$ ns
  - Turn-off time: $t_{off} = 70$ ns
  - Duty ratio: $D = 0.6$
- Assume the drain current is constant at 10A and $V_{DS}$and $I_D$change linearly.
- Calculate: (a) The conduction loss $P_{\text{cond}}$of the MOSFET. (b) The turn-on switching loss P_{on}. (c) The turn-off switching loss P_{off}. (d) The total switching loss P_{sw}. (e) The total MOSFET power loss.

Example

- Calculate: (a) The conduction loss P_{cond} of the MOSFET. (b) The turn-on switching loss P_{on}. (c) The turn-off switching loss P_{off}. (d) The total switching loss P_{sw}. (e) The total MOSFET power loss.
- $P_{\text{cond}} = I^2_{\text{MOSFET,rms}} R_{DS(on)}$.
- Therefore, over one switching period, $I_{\text{MOSFET,rms}} = I_D\sqrt{D}$
- For $I_D = 10\text{A}$ and $D = 0.6$, $I_{\text{MOSFET,rms}} = 10\sqrt{0.6} = 7.746$ A
- Hence, $P_{\text{cond}} = 2.4\ W$

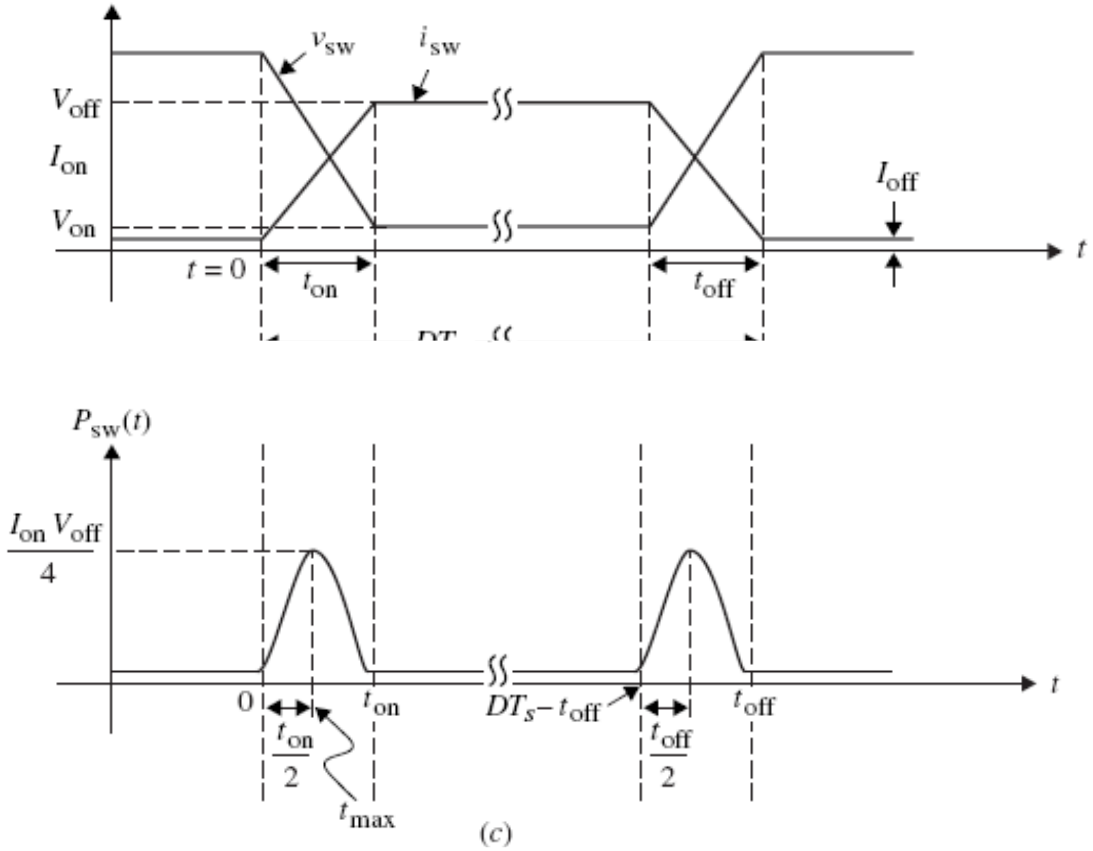
Example

- Calculate: (a) The conduction loss P_{cond} of the MOSFET. (b) The turn-on switching loss P_{on}. (c) The turn-off switching loss P_{off}. (d) The total switching loss P_{sw}. (e) The total MOSFET power loss.
- We have seen: $E_{on} = \frac{1}{6} V_{DS} I_D t_{on}$
- $t_{on} = 50$ ns $= 50 \times 10^{-9}$ s, therefore, $E_{on} = \frac{1}{6}(48)(10)\left(50 \times 10^{-9}\right) = 4.0 \times 10^{-6}$ J= 4 μJ
- The turn-on switching power is
- $P_{on} = E_{on} f_s = \left(4 \times 10^{-6}\right)\left(100 \times 10^{3}\right)$
- Pon=0.4 W

Example

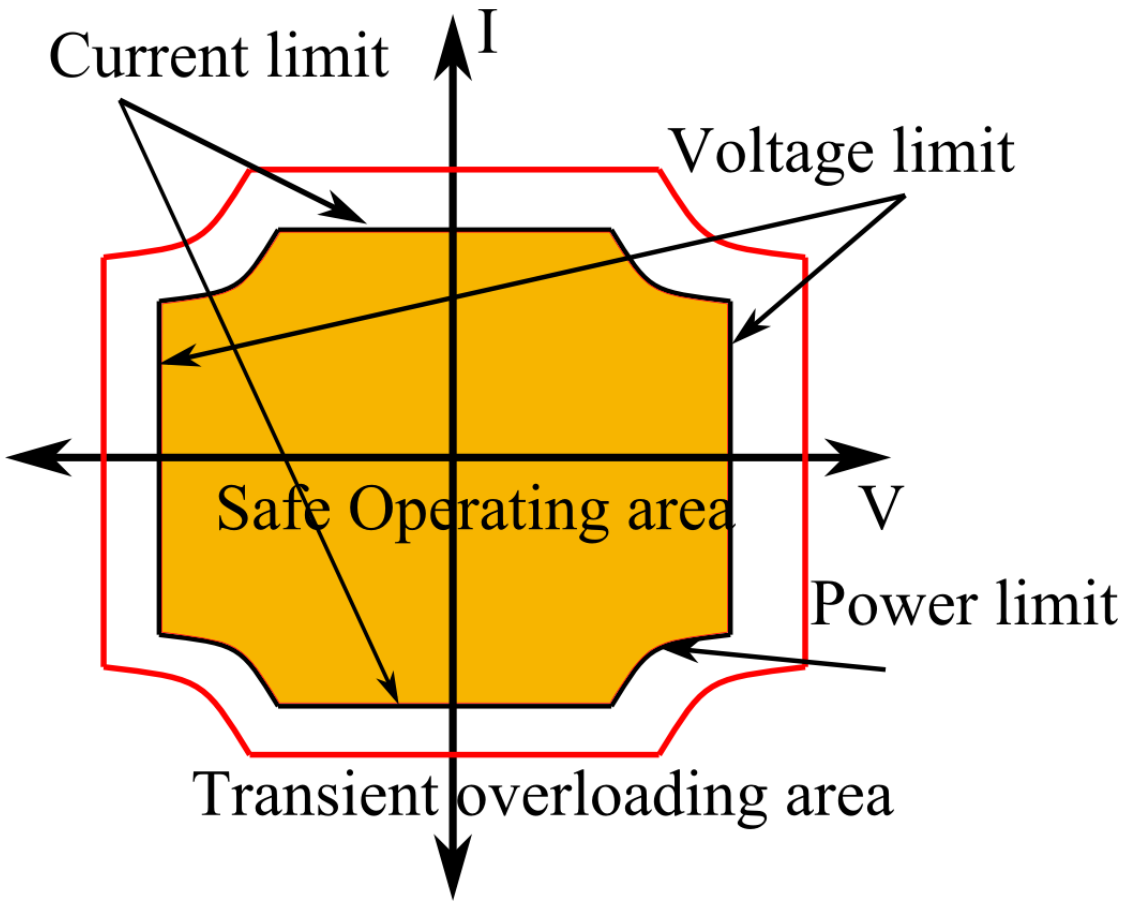
- Calculate: (a) The conduction loss P_{cond} of the MOSFET. (b) The turn-on switching loss P_{on}. (c) The turn-off switching loss P_{off}. (d) The total switching loss P_{sw}. (e) The total MOSFET power loss.
- Similarly, $E_{off} = \frac{1}{6}(48)(10)\left(70 \times 10^{-9}\right) = 5.6\,\mu\text{J}$
- The corresponding power loss is $P_{off} = E_{off} f_s$ =0.56 W
- Hence, total switching loss: $P_{sw} = P_{on} + P_{off} = 0.4 + 0.56 = 0.96$ W
- The total loss consists of conduction and switching losses:
- $P_{\text{MOSFET}} = P_{\text{cond}} + P_{\text{sw}} = 2.4 + 0.96 = 3.36$ W

Do it yourself



- Show that the peak power dissipated during switching is given by
- $P_{peak} = \frac{I_{on}V_{off}}{4}$
- Why is this quantity important?

Operating area of a switch



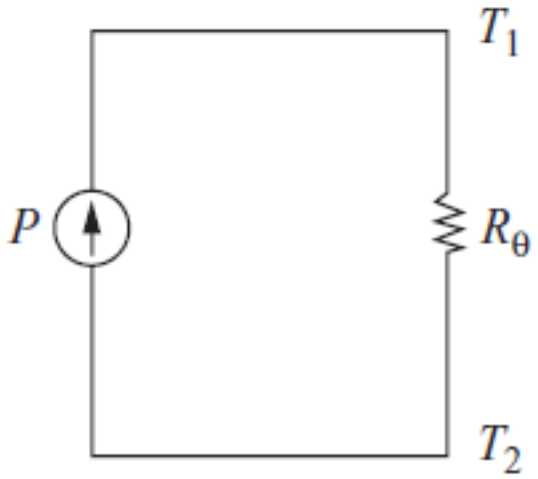
- Steady state safe operating area of a device is shown with voltage, current and power limits.
- Short time transient overloaded area is slightly bigger in size.
- Each power device has different safe operating area.

Limits of a switch

- Apart from voltage and current, thermal limit is the most important limiting factor for a power switch.
- It is also true for other components.
- Temperature rise should always be kept much below the maximum permissible limits for long lifetime of the switch.
- Every 10 degree C rise in temperature reduces the device lifetime by half.
- Heat sinks help to reduce the temperature of the power device by increasing the surface area of contact with the surrounding.
- To reduce the temperature further, fans may be used.

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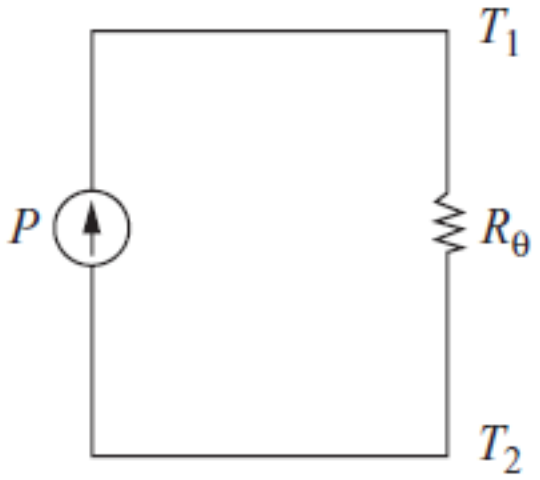
Thermal model



- The temperature difference between two points is a function of the thermal power generated and the thermal resistance between the two points.
- Thermal resistance is $R_\theta = (T_1-T_2)/P$

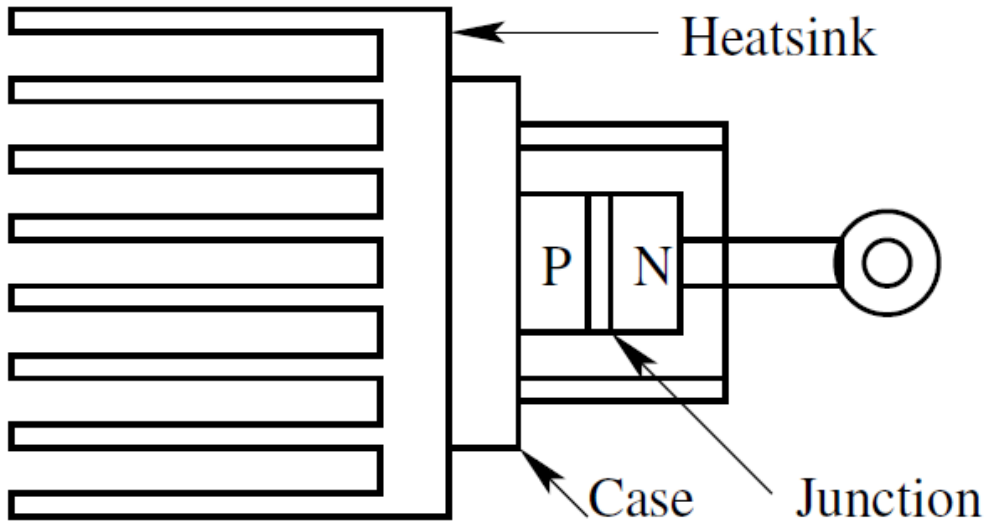
  where (T_1-T_2) is the temperature difference (K), P is the thermal power (W) and R_θ is the thermal resistance (K/W).

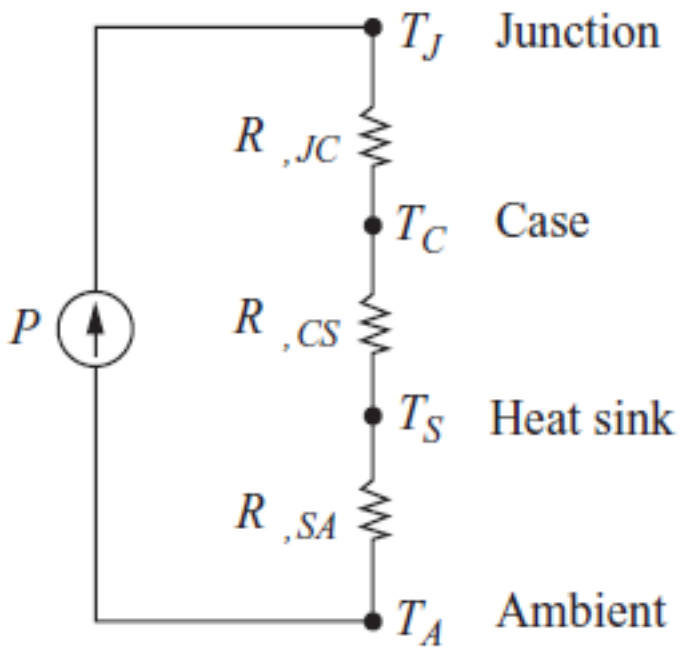
Thermal model



- An electrical equivalent circuit is shown.
- If T2 is the temperature of ambient and T1 is the temperature of the junction, then a heat sink will reduce R_θ between the junction and the ambient, and hence T1 will decrease.
- A heat sink with a fan will further reduce R_θ .

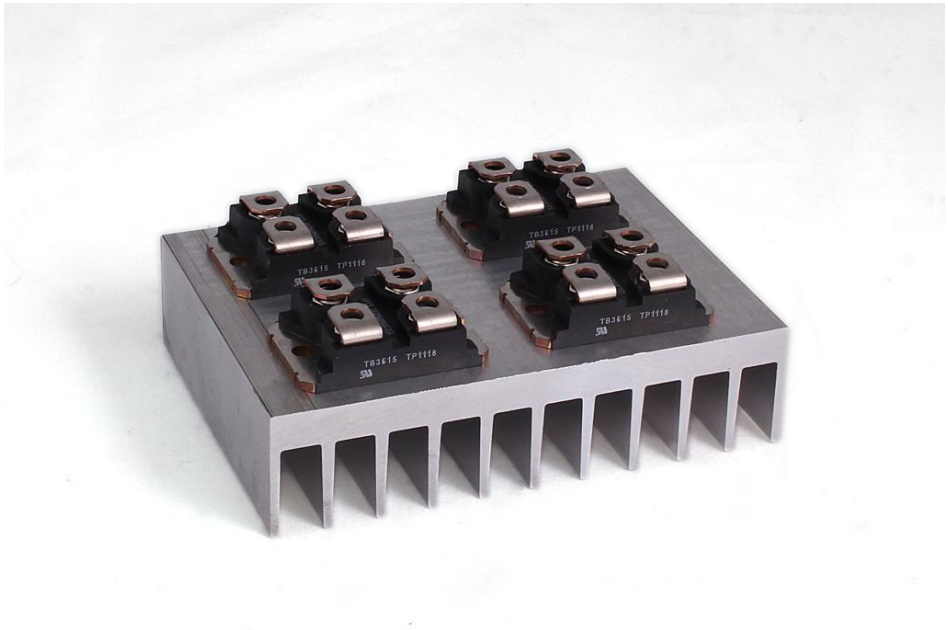
Thermal model





- Thermal power flows from junction to case, then from case to heatsink, then from heat sink to ambient.
- This is modelled as three series resistances: $R_{\theta JC}$, $R_{\theta CS}$, $R_{\theta SA.}$ Semiconductor and heat sink manufacturers provide these values.

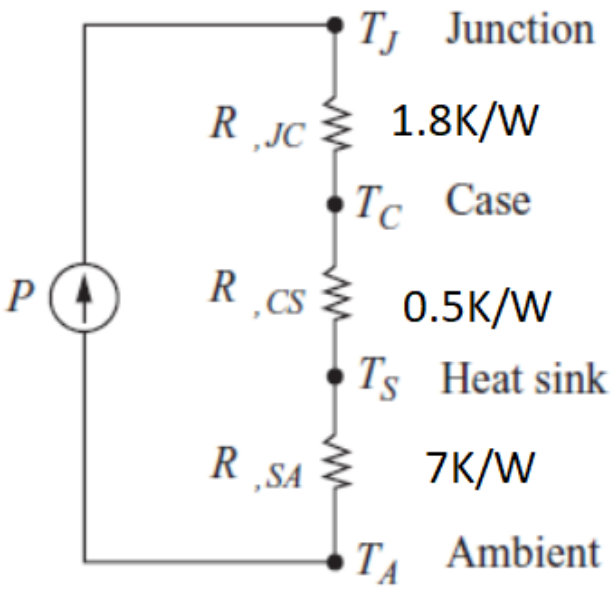
Images of heat sinks





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Example of heat sink selection



- Consider a MOSFET whose junction to case thermal resistance is 1.8K/W and case to heat sink thermal resistance is 0.5K/W. If the device is mounted on a heat sink that has a thermal resistance of 7K/W, then what is the maximum power that can be allowed to be dissipated in the MOSFET ? Given, maximum junction temperature of 150^0C and ambient temperature 40^0C.
  - P_{max}= (150-40)/(1.8+0.5+7) = 11.8 W
- The switching frequency of the MOSFET is now increased and the loss is increased to 15W. What will be the junction temperature now?
  - Junction temperature= 15x(1.8+0.5+7)+40= 179.5^0C. Not acceptable.
- Determine the thermal resistance of a new heatsink that will limit the junction temperature to 150^0C.
  - $R_{heatsink}$= (150-40)/(15) – 1.8 - 0.5 = 5.03 K/W.

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